

Teacher Guide — primary / module-05-dot-addition

IKS Curriculum

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Overview

Module 5 — Make Ten! (Towards Bindu Paddhati)

Age band: Primary · **Grades:** 3–5 · **Duration:** 10 sessions × 35 min

What children will learn

1. The five 10-pairs: 1+9, 2+8, 3+7, 4+6, 5+5. Instantly.
2. Spot 10-pairs in any list of digits.
3. Add a small column of single digits (4–6 numbers) by grouping into 10s.
4. Mental shortcut: “add a 9” = “add 10 and take 1 away.”
5. A peek at the dot method (*Bindu Paddhati*) as a “trick the older kids will learn.”

How this module sounds

- Lots of games. Cards. Dice. Chanting.
- No long worksheets. Children move, talk, count out loud.
- Sanskrit term *Bindu Paddhati* introduced once on Day 6 as “the dot method name in Sanskrit.”

NEP 2020 alignment

- §5.1 — Foundational numeracy
- §4.6 — Indian Knowledge Systems
- §4.23 — Experiential learning

Materials

- A deck of number cards 0–9 (4 sets per pair)
- Dice (one per pair)
- “Make 10” wall poster
- Counters (beads, buttons)
- Drawing paper

Day-by-day

Day	Title	What we do
1	Make Ten Game	Card game with pairs
2	Bonds of Ten chant	Memorise the 5 pairs
3	Find the Pair (in lists)	Scan + circle
4	Add by Grouping	Small column drill via game
5	The Add-9 Trick	Add 10, take 1
6	Meet the Dot (Bindu Paddhati)	Brief demo by teacher
7	Make-a-Game Day	Children design their own card game
8	Project Day 1	Start their own “make 10” card or board game
9	Project Day 2	Finish + practise
10	Game Showcase	Play each other’s games

Project

Children design their own “make 10” card or board game in pairs. They play each other’s on Day 10.

See [assessments/project-brief.md](#).

Sources

The “make 10” pattern is foundational arithmetic — every culture’s basic teaching includes it. The Indian “dot method” name (*Bindu Paddhati*) comes from 20th-century Indian math pedagogy. For older grades’ deeper treatment, see the Middle-band version of this module.

10-Day Lesson Plan

Module 5 (Primary) — 10-Day Lesson Plan

At a glance (35-min sessions)

Day	Title	Activity	Output
1	Make Ten Game	Pair card game	First-day fun
2	Bonds of Ten	Memorise pairs	Chant + recall
3	Find the Pair	Scanning drill	Worksheet
4	Add by Grouping	Column-addition	First column wins

Day	Title	Activity	Output
		game	
5	Add-9 Trick	Add 10, take 1	Drill sheet
6	Meet the Dot	Teacher demo of Bindu Paddhati	Awe
7	Make-a-Game Day	Design own game	Sketch + first prototype
8	Project Day 1	Build the game	Game ready to play
9	Project Day 2	Finish + test	Play once
10	Game Showcase	Play each other's games	Stickers

Daily structure (35 min)

Time	What
5 min	Settle + the “Make 10!” chant
8 min	Story / new game introduction
17 min	Game / activity
5 min	Sharing / reset

The daily chant

“1 and 9 make 10! 2 and 8 make 10! 3 and 7 make 10! 4 and 6 make 10! 5 and 5 make 10!”

Sing it / chorus it with claps. By Day 3 it should be effortless.

Pacing notes

- Games > worksheets at this band. If a worksheet feels slow, drop it and add another game round.
- Day 6 (dot method demo) is a “wow” moment — keep it short and mysterious.
- Days 8–10 build to the game showcase. Confirm groups by Day 6.

What children leave with

- Instant recall of all 5 ten-pairs.
- A mental habit of looking for “easy groups” when adding.
- The Add-9 trick.
- Curiosity about the dot method.
- One game they made and one they played.

Teacher Notes

Teacher Notes — Module 5 Dot Addition (Primary)

Posture

Make-10 fluency is the single skill that pays off in every later math module. This is what we're building. The dot method itself is a Middle-band topic; we *peek* at it on Day 6 to plant the seed.

Common things to handle

1. **A child can't yet recall all the 10-pairs.** That's fine — the module is the practice. Don't shame; bring them into more games.
2. **A child gets bored ("I already know this!").** Give them the "scanning" challenge: find ALL 10-pairs in a list of 8 digits in 5 seconds.
3. **A child says "1+9 isn't always 10, what about negative numbers?"** Acknowledge — *"That's a big question for older grades. For now we're working with the counting numbers."*

Safety

- No physical risk in this module.

Differentiation

- **Tier up:** Grade 5 children may be ready for 2-digit + 1-digit mental addition using the Add-9 trick. Push to $47+9 = 56$ by Day 5.
- **Tier down:** Grade 3 children still building 10-pair fluency need EXTRA game time. Skip the column-addition game (Day 4) and do another 10-pair game instead.

Project oversight

- Days 7–10: children design a "make 10" game. Encourage variety — card games, board games, dice games, jumping games.
- Materials limit: only what's in the classroom. Don't expect families to provide game supplies.

If you only have 6 days

- Day 1: Make Ten Game
- Day 2: Bonds of Ten chant + Find the Pair
- Day 3: Add-9 Trick
- Day 4: Meet the Dot + Game Day
- Day 5: Project Day
- Day 6: Game Showcase

Skip the dedicated column-addition game.

Sources

Sources — Module 5 (Primary)

At this band, sources are teacher reference only.

Source for the 10-pair pedagogy

- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center) — §5.2 Mental Addition. Available in `/Users/sree/Projects/SutraGanita/content/`.
- The dot method itself is in §5.4 of the same source.

Why we don't teach the full dot method at Primary

The dot method requires: - Fluent 10-pair recall (foundational) - Scanning a column for groups (intermediate) - Place-value understanding for the carry (intermediate)

Most Grade 3–4 children are still building the first. We focus there and tease the rest.

For families who ask

The Bindu Paddhati (“dot method”) is a mental-arithmetic technique taught in Indian schools. We introduce its building blocks now; the full method comes in Grade 6–8.

Day 1 — Make Ten Game

Module: Make Ten / Bindu Paddhati (Primary) · **Time:** 35 min

Goal

Children play a card game that requires finding pairs that sum to 10.

Materials

- A deck of number cards 0–9 (4 of each = 40 cards) per pair
- Counters (optional)

The flow

Settle + introduce the chant (5 min)

Teach the chant — clap on each pair:

“1 and 9 make 10! 2 and 8 make 10! 3 and 7 make 10! 4 and 6 make 10! 5 and 5 make 10!”

Say it 3 times.

Story (5 min)

“Today we play a game called Make Ten. The rule is simple: find two cards that add up to 10. The player with the most 10-pairs wins.”

Game (20 min)

How to play:

1. Shuffle the deck. Deal 5 cards to each player. Place 5 cards face-up in the middle.
2. On your turn:
 - Look at your hand and the middle cards.
 - Find a pair that makes 10 (one from your hand + one from the middle).
 - Take the pair. Put it in your “pile.”
 - Draw a new card from the deck.
3. If you can’t make 10, place one card in the middle and end your turn.
4. When the deck runs out, count pairs. Most pairs wins!

Play 2–3 rounds.

Show + chant (5 min)

- “Who won a round?” — winner shares ONE 10-pair they made.
- Class chants the pair: “6 and 4 make 10!”
- Close with the daily chant.

Homework

- At home, practise the chant with a sibling or parent. Win a round of Make Ten with family!

Differentiation

- **Tier up:** Play with 0–9 cards including 0+10 pairs.
- **Tier down:** Use only cards 1–6 to keep the pairs simple.

Teacher’s quick notes

- Some children won’t find pairs and will give up. Pair them with a stronger player as a “team.”
- The game IS the lesson at this band. Don’t add a worksheet.

Day 2 — Bonds of Ten Chant

Module: Make Ten (Primary) · **Time:** 35 min

Goal

Children memorise the five 10-pairs through chant, hand games, and repeated quick-recall.

Materials

- Wall poster of the 5 pairs (with pictures)
- Number cards 0–9

The flow

Settle + chant (5 min)

Daily chant: “*1 and 9 make 10!...*”

The hand game (10 min)

- Children pair up.
- Partner A holds up some fingers (1–9).
- Partner B holds up the rest to make 10.
- Check: do their fingers together = 10?
- Swap, repeat.

Play for 5 min. Then teacher leads class: - Teacher holds up 6 fingers. Class chorus: “FOUR!” - Teacher holds up 8. Class: “TWO!” - Mix it up. Faster.

Walk the pairs (10 min)

- 5 stations around the room, one per pair (1+9, 2+8, 3+7, 4+6, 5+5).
- Each station has a picture + the equation.
- Children walk to each station and say the pair aloud.

Quick-fire recall (8 min)

Teacher holds up a number card. Class shouts the partner.

- 5 → “FIVE!” (5+5)
- 7 → “THREE!”
- 9 → “ONE!”
- etc.

Do 20 cards in 60 seconds. Then do it again to beat the time.

Show + chant (2 min)

Close with the daily chant — bigger than ever.

Homework

- At home tonight: parent or sibling holds up some fingers. You give the rest to make 10. Win 10 in a row!

Differentiation

- **Tier up:** Find pairs that make 11 too (5+6, 4+7).

- **Tier down:** Use a “make-10 chart” if recall is slow. Withdraw the chart by Day 5.

Teacher’s quick notes

- This is a memorisation day. Repetition over depth.
- The hand-game pairings often surface friendship dynamics — be flexible.

Day 3 — Find the Pair (Scanning Drill)

Module: Make Ten (Primary) · **Time:** 35 min

Goal

Children spot 10-pairs in a list of 6–8 single-digit numbers within 10 seconds.

Materials

- Worksheet with 10 rows, each a list of 6–8 digits (printable)
- Pencils

The flow

Settle + chant (5 min)

Daily chant.

Demonstration (5 min)

- Teacher writes on the board:
3 7 2 5 8 1 4 6
- “*Find ALL the pairs that make 10. Circle them.*”
- Walk through together: 3+7, 2+8, 4+6. Three pairs. Leftover: 5, 1.

Scanning drill (20 min)

- Hand out worksheet.
- Individual work: circle every 10-pair in each row. Note the leftover digits.
- 10 rows. About 1 minute per row.

After 10 min, **paired check:** partner reads a row aloud; you find pairs in under 10 seconds.

Speed round (5 min)

Teacher projects one row at a time on the board:

4 6 2 8 3 7
1 9 5 5 3 7
8 2 6 4 7 3

Class shouts pairs in chorus. 20 seconds per row.

Show + chant (close)

Daily chant.

Homework

- 5 more scanning rows on the back of the worksheet.

Differentiation

- **Tier up:** Find pairs that make 10 AND pairs that make 11.
- **Tier down:** Work with shorter rows (4 digits each).

Teacher's quick notes

- The pattern of “scan and group” is the key skill — more important than getting every pair right.
- If a child circles $5+5$ as one pair, that's correct (both 5s used together).

Day 4 — Add by Grouping

Module: Make Ten (Primary) · **Time:** 35 min

Goal

Children add a column of 4–6 single-digit numbers by first grouping into 10s, then adding leftovers.

Materials

- Dice (one per pair)
- Worksheet with 5 small columns
- Counters

The flow

Settle + chant (5 min)

Daily chant.

Demonstration (8 min)

Write on the board:

6
3
7
5

4
2

“Today we add a column. But not the slow way. The smart way.”

Walk through: - Look for pairs first. $6+4 = 10$. (Cross out the 6 and the 4.) - $3+7 = 10$. (Cross out the 3 and the 7.) - That leaves 5 and $2 = 7$. - So total = $10 + 10 + 7 = 27$.

Compare: the slow way would be $6+3=9$, $+7=16$, $+5=21$, $+4=25$, $+2=27$. Same answer, more effort.

Dice game (15 min)

- Pairs play.
- Roll the die 5 times. Write the 5 numbers in a column.
- Both players race to find the answer using the grouping trick.
- First with correct answer = 1 point.
- 10 rounds. Most points wins.

Worksheet (5 min, individual)

5 small columns. Find groups, write the answer.

Show + chant (2 min)

Close.

Homework

- 5 more dice rolls at home. Practise the trick with family.

Differentiation

- **Tier up:** Columns of 7–8 numbers; find as many 10-groups as possible.
- **Tier down:** Columns of 3–4 numbers; just find ONE pair.

Teacher’s quick notes

- The dice game introduces healthy competition with low stakes.
- Watch for children who add without looking for pairs first — gently steer them back to “look first.”

Day 5 — The Add-9 Trick

Module: Make Ten (Primary) · **Time:** 35 min

Goal

Children add 9 to any 2-digit number using the trick “add 10, take 1 away.”

Materials

- Whiteboard
- “Add 9” drill sheet
- Quick-fire flash cards

The flow

Settle + chant (5 min)

Story / trick (10 min)

- Write on the board:

$$33 + 9 = ?$$

- “*Anyone want to try?*” Take 2 answers. (Usually 42 with some scratch-out work.)
- “*What if I told you there’s a trick? Watch:*”

9 is the same as $10 - 1$.

So $33 + 9$ is the same as $33 + 10 - 1$.

$$33 + 10 = 43.$$

$$43 - 1 = 42.$$

Done!

“Add 10, take 1 away. Easy.”

More examples: $- 47 + 9 = 47 + 10 - 1 = 56 - 64 + 9 = 64 + 10 - 1 = 73 - 28 + 9 = 28 + 10 - 1 = 37$

Drill (15 min)

Worksheet of 20 “+9” problems. Or quick-fire flash cards.

After 5 min of individual: paired race — who can do 5 in 30 seconds?

Speed round (3 min)

Teacher calls “27 plus 9” — class shouts “36!” Mix it up.

Show + chant (2 min)

Close.

Homework

- Find ANY 5 2-digit numbers (look at your house number, car plate, telephone digits). Add 9 to each in your head.

Differentiation

- **Tier up:** Try the trick for adding 19 (add 20, take 1 away). $47 + 19 = 66$.

- **Tier down:** Stick to +9 only. Build up.

Teacher's quick notes

- The trick generalises to +99, +999 too — but at this band stay with +9.
- Some children will resist: “but I just add!” Honour that — say “you have two tools now, use whichever’s faster.”

Day 6 — Meet the Dot (Bindu Paddhati)

Module: Make Ten (Primary) · **Time:** 35 min

Goal

Children see a demonstration of the dot method (*Bindu Paddhati*) — the technique they’ll learn fully in Grade 6+.

Materials

- Whiteboard, large
- A long column problem (4 four-digit numbers)

The flow

Settle + chant (5 min)

Story (8 min)

- “*Today I’m going to show you a magic trick. Older kids — Grade 6, 7, 8 — learn this. You’re going to SEE it today. We won’t fully learn it until you’re older.*”
- “*It’s called Bindu Paddhati — that’s Sanskrit for ‘dot method.’*”

Live demo (12 min)

Write a 4-row column:

5728
3417
6892
4536+

Solve LIVE with the dot method, narrating: - Units column: 8, 7, 2, 6. - $8+2 = 10$! Put a dot above the 2. - Now I have 7 and 6 left. $7+6 = 13$. That’s a 10! Dot above the 6. - 3 is left over. Write 3 below. - I have 2 dots. Carry 2. - Tens column: 2, 1, 9, 3 + the carry 2. - $1+9 = 10$. Dot above 9. - $2+3+2$ (carry) = 7. - Write 7. Carry 1. - Hundreds: 7, 4, 8, 5 + 1. - Continue. - Thousands.

Reveal: **20573**.

“*How long did that take? About a minute. Try doing this the slow way — much longer.*”

Awe + connect (8 min)

- *“You already know the trick of finding 10-pairs. That’s the BIG SKILL behind this magic. When you’re in Grade 6, you’ll learn the dot method too. For now, keep practising those 10-pairs.”*
- Class chorus of all 5 pairs.

Show + chant (2 min)

Close.

Homework

- Teach the chant to a younger child at home. They’ll be ready when they’re older.

Differentiation

- **Tier up:** Grade 5 fast learners can try one small column problem (4 single digits) with the dot method.
- **Tier down:** Just watch the demo. Today is awe-day, not practice-day.

Teacher’s quick notes

- The “magic” is the point. Don’t over-explain. Let it be mysterious.
- Don’t push for understanding — just curiosity.

Day 7 — Make-a-Game Day

Module: Make Ten (Primary) · **Time:** 35 min

Goal

In pairs, children design (paper sketch + first prototype) their own “Make 10” game.

Materials

- Drawing paper, markers
- Spare cards, dice, counters

The flow

Settle + chant (5 min)

Inspiration (5 min)

- *“You’ve played lots of games this week. Now YOU design one. Your game must include the ‘Make 10’ idea — somehow.”*
- Quick brainstorm. Take 5 student ideas. Write on board.

Examples: - A card game where you collect pairs - A board game with “Make 10” squares - A jumping game where you jump to the number that makes 10 - A hide-and-seek game with number cards

Design + sketch (15 min)

- Pairs decide on ONE game.
- Sketch the game on paper: rules, what you need, how to win.
- Make a quick prototype if possible.

Test it (8 min)

- Each pair tries to play their own game once. Notice what works and what doesn't.

Show + chant (2 min)

- 2 pairs share their game idea aloud.
- Daily chant.

Homework

- Finish your game design at home tonight. Bring all materials tomorrow!

Differentiation

- **Tier up:** Add a “twist” — what makes your game extra fun or tricky?
- **Tier down:** Adapt an existing game (snakes and ladders, but every step you say a 10-pair).

Teacher's quick notes

- This is creative day. Don't over-direct. Wild ideas welcome.
- If a pair is totally stuck, suggest: “modify Snakes and Ladders so every ladder requires a Make 10 answer.”

Day 8 — Project Day 1

Module: Make Ten (Primary) · **Time:** 35 min

Goal

Each pair builds a playable version of their game.

Materials

- Cards, dice, drawing paper, markers, glue, scissors

The flow

Settle + chant (5 min)

Project work (25 min)

- Pairs work to build their game:
 - Make the cards / board
 - Write the rules clearly (so another pair can read and play)
 - Decorate if there's time

Teacher rotates: 60 seconds per pair, ONE question: *“Can another pair play your game with just the rules?”*

Share progress (3 min)

2 pairs hold up their game.

Show + chant (2 min)

Daily chant.

Homework

- Finish at home what you didn't finish in class. Tomorrow we play!

Differentiation

- **Tier up:** Add a “second round” or “challenge” level to your game.
- **Tier down:** Keep it simple — fewer cards, fewer rules.

Teacher's quick notes

- The “rules clear enough for another pair” criterion is the actual project standard.
- Spend extra time on pairs whose game has unclear rules — most common issue.

Day 9 — Project Day 2

Module: Make Ten (Primary) · **Time:** 35 min

Goal

Each pair finishes their game and tests it with another pair.

Materials

- All in-progress games

The flow

Settle + chant (5 min)

Final touches (15 min)

Pairs complete their game. Rules card, board, all pieces ready.

Test swap (12 min)

- Pair up with a buddy pair. Swap games.
- Read each other's rules. Play their game for 5 minutes.
- Then give feedback: 1 thing that worked, 1 thing that was confusing.

Show + chant (3 min)

- 2 pairs share what they learned from testing.
- Daily chant.

Homework

- Make any final fixes. Bring everything tomorrow!

Differentiation

- **Tier up:** Add a sticker / reward system to your game design.
- **Tier down:** Simpler rules. Less is more.

Teacher's quick notes

- The test swap is the real test of clarity. Pairs whose game is hard to follow now have evidence.
- Help pairs who realise their rules are too complex — encourage them to *simplify* not redesign.

Day 10 — Game Showcase

Module: Make Ten (Primary) · **Time:** 35 min

Goal

The whole class plays each other's games in a rotation. Every child gets a sticker.

Materials

- All children's games set up at stations
- Stickers

The flow

Settle + chant (3 min)

The biggest chant yet.

Game stations (25 min)

- Each pair sets up their game at a station.
- Class rotates: 5 minutes per game station, in groups of 4.
- One member of the original pair stays at the station to teach the rules; the other plays others' games.

(Adjust based on class size — with 10 pairs and 5 minutes per station, plan 4–5 station rotations.)

Closing ceremony (7 min)

- Whole class together.
- “Best game I played today” — quick share by 5 children.
- Stickers all round.
- Final chant — loudest yet.

Show + chant

Done!

Homework

- None. Module complete.

Differentiation

- **Tier up:** A confident pair can teach two stations at once.
- **Tier down:** A shy pair can teach with just one card of rules; teacher helps.

Teacher's quick notes

- This is celebration day. No failure. Every pair's game is a win.
- Take photos of the games and rules; send to parents.
- The skill of “rules clear enough to teach a stranger” is a real life skill — name it explicitly to the class.

Oral Recall — Module 5 (Primary)

No graded paper tests. Use these oral checks.

Day 2 — 10-pair recall

Teacher holds up a digit. Class chorus the partner. Do 10 in 30 seconds.

$8 \rightarrow 2 / 6 \rightarrow 4 / 3 \rightarrow 7 / 5 \rightarrow 5 / 9 \rightarrow 1 / 7 \rightarrow 3 / 4 \rightarrow 6 / 2 \rightarrow 8 / 1 \rightarrow 9 / 0 \rightarrow 10$

Note: any child needing prompting works with you in next class's settle.

Day 4 — Grouping check

Show column on board:

6, 3, 4, 7

Class shouts: "6+4=10! 3+7=10! Total = 20!"

Day 5 — Add-9 quick fire

Teacher calls "27 plus 9!" — class shouts "36!"

10 problems in 60 seconds.

Day 10 — Game completion sticker

Every child plays at least one game = sticker. No further assessment.

Optional written check

Match the pair (Day 3 or 5):

1	6
2	5
3	4
4	3
5	2
6	1

Should be $1 \rightarrow 9$, $2 \rightarrow 8$ etc. for a Make-10 version. Adapt as needed.

Activity 01 — Make Ten Card Game

Module: Make Ten (Primary) · **Day:** 1 (and reusable) · **Time:** 20 min within Day 1

Goal

Children play a paired card game finding pairs that sum to 10.

Materials per pair

- One deck of 40 number cards (0–9, 4 of each)

Set-up

- One deck per pair on the desk.

Rules (child-facing)

Make Ten — for 2 players.

1. Shuffle. Deal 5 cards to each player. Put 5 face-up in the middle.
2. On your turn:
 - Look at your hand + the middle.
 - Find 2 cards that add to 10 (one from your hand + one from middle).
 - Take both. Put in your pile.
 - Draw 1 from the deck.
3. If you can't make 10, put 1 card in the middle. End your turn.
4. When the deck is empty, count your pairs. Most pairs wins!

Going well

	Looks like	Sounds like
Working	Both children scanning cards, trading positions, claiming pairs	"I got 6 and 4!" "Your turn!"
Stuck	Long silences, one player taking all turns	"I don't know" / no eye contact

Differentiation

- **Tier up:** Use 0–9; pairs of 10 only.
- **Tier down:** Use 1–6; pair sums of 10 are rarer, allow pair sums of 7 too.

Clean-up

- Cards back in the deck. Stack on the teacher's desk.

Project Brief

Project — Module 5 (Primary)

What you'll make

In pairs, design and build a "Make 10" game. Anything goes: cards, board game, dice game, jumping game. The only rule: somewhere in your game, players have to find or use pairs that add to 10.

Schedule

- **Day 7:** sketch your game.
- **Day 8:** build it.
- **Day 9:** test with a buddy pair.

- **Day 10:** play each other's games at the Showcase.

What we'll celebrate (simplified rubric)

	What we celebrate
	Your game has the "Make 10" idea in it
	The rules are clear enough that another pair can read and play
	Your game is fun to play — at least one buddy-pair-member smiled

All three stars = sticker + photo for parents.